



interfaceNode - *Datasheet*

Version 1.0.6

Hardware Revision: B.0.x

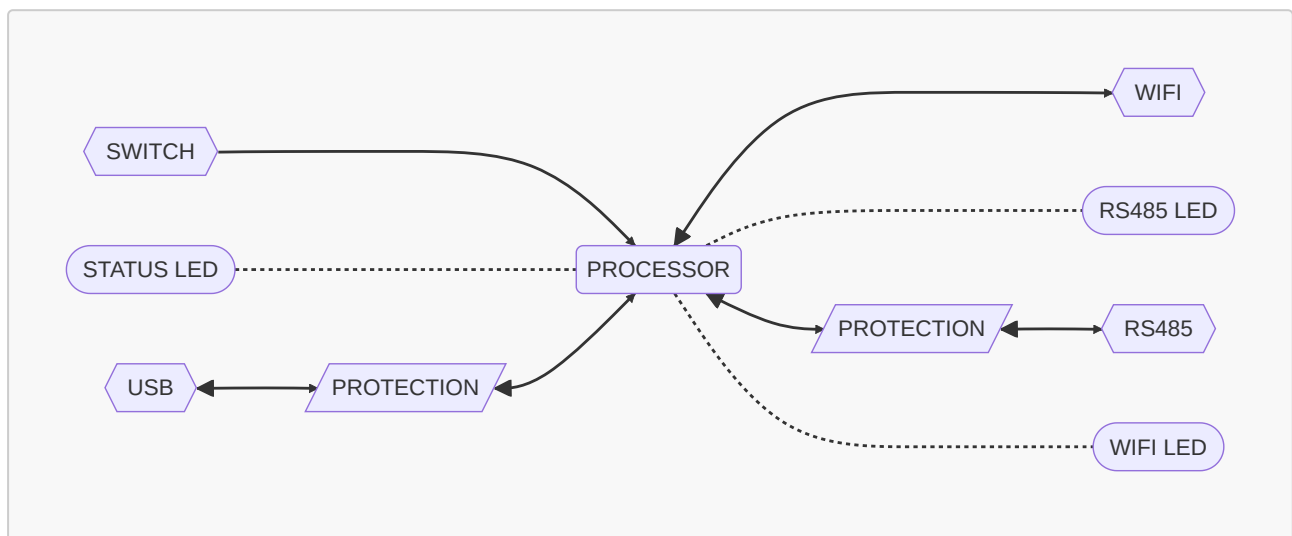
The Atkin Engineering **interfaceNode** is a Wireless RS-485 bridge that provides bridging connections between RS-485, USB, TCP sockets, and WebSockets for various control devices.

front

rear



Block Diagram



Specifications

Mechanical

- Dimensions 120 x 50 x 30 mm (4.72 x 2.36 x 1.18 in)
- Weight 105 grams (2.71 oz)

Electrical

- 12/24 VDC (absolute 5 - 42 VDC) from RJ12 port
- 1.2 W average (2 W (peak)) [^current]
- Input Protection to ± 42 VDC

RS485

- IEC 61000-4-2 Level 4 ESD (± 8 kV contact, ± 15 kV air); IEC 61000-4-4 Level 4 EFT (2 kV); IEC 61000-4-5 Level 2 surge immunity (4 kV); ± 70 VDC bus-fault tolerance.

USB

- USB 2.0 full-speed CDC; 2.5 kV isolation; 5 kV transient/surge immunity.

Services

RS485

- Full / Half duplex [^jumpers]
- Termination **None** / **120ohm** [^jumpers]
- Protocol aware (DyNet 1 & 2, or raw passthrough)
- Baud rates: 9600 / 19200 / 38400 / 57600 / 115200

WiFi (Access Point / Station)

- IEEE 802.11b/g/n-compliant
- Supports 20 MHz and 40 MHz bandwidth in 2.4 GHz band
- 1T1R mode with data rate up to 150 Mbps
- Center frequency range of operating channel: 2412 ~ 2484 MHz
- WPA2/WPA3 Personal; WPA2/WPA3 Enterprise (802.1X) supported in Station mode
- Station mode falls back to the configuration access point if the network can't be reached after 5 attempts (about two minutes), with the reason recorded in the system log

Bridge

- Bridges data between four interfaces: USB CDC, RS485, TCP sockets (up to 5 clients), and WebSockets (up to 5 clients, standard ports: **wss://** 443, **ws://** 80)
- Protocol aware (DyNet 1 & 2, checksum-validated) or raw passthrough - applies across all four interfaces
- WebSocket traffic can be filtered by DyNet frame type and by opcode, configurable from the web interface
- DyNet discovery beacon (UDP port 9998) and mDNS device discovery - only running services are advertised, under standard service types that follow their TLS setting (**_https/_http**, **_ssl/_socket**, **_wss/_ws**)

Security

- Unique default password for every device, **XXXX-XXXX-XXXX** on the label - no shared factory password
- Login on every interface: web interface (session login), WebSocket bridge (one-time token, PDDEG-S style) and TCP bridge (login line); repeated wrong passwords lock the login for an increasing time

- Encryption with TLS 1.2 (ECDHE key exchange, AES-GCM) for the web interface (HTTPS), WebSocket bridge (WSS) and TCP bridge - on by default, each can be turned off for clients that can't use it
- Per-device certificate authority: install its certificate once per computer and every app trusts the device, even when its IP address changes (ECDSA P-256 keys made on the device); your own certificate can be uploaded instead
- Passwords stored only as salted hashes (PBKDF2-HMAC-SHA256); settings storage (NVS) encrypted
- Full factory reset erases all user data, credentials and the device certificates, then starts in configuration mode ready for setup
- Cross-origin (CORS) access for web apps can be turned off
- Security event log (logins, lockouts, setting changes, firmware updates, factory resets, restart reasons) with date and time, the device clock being set from the browser at each login; opt-out available; no passwords or keys recorded
- Web page shows the security state at a glance: green, amber or red page border for secured, login only, or open services
- Watchdog restarts the device if a data task stops responding

Configuration & Firmware

- Web-based configuration over HTTPS, no software installation required
- USB drive (6 MB) in configuration mode for copying files to and from the device
- System log viewable from the web interface
- Firmware update over WiFi from the web interface, with automatic rollback if the new firmware fails to start

Support and Security Updates

- Security updates are provided free of charge for 7 years from each unit's production date. The end date (month and year) is printed on the product label and box.
- Security fixes released as patch versions (x.y.Z), separately from feature releases, on the [Releases](#) page
- Each release includes a software bill of materials (SBOM) and vulnerability report
- Vulnerability reporting: see [SECURITY.md](#) in the firmware repository

Connections

Front

- USB Type-C, USB 2.0 full-speed port [[^]usbpower]
- 1x Status LED
- 1x System button

Rear

- 2x RS-485 (parallel RJ12 6P6 ports) [[^]rs485power]
- 2x RGB status LEDs

Environment

- Operating Temperature: 5° to 40°C ambient (41° to 104°F)
- Storage/Transport Temperature -20° to 65°C ambient (-4° to 149°F)
- Humidity 0 to 90% non-condensing

Compliance

- EU: CE - Radio Equipment Directive 2014/53/EU, including Article 3(3)(d) and (e) (EN 18031-1, EN 18031-2, notified body EU-type examination); RoHS 2011/65/EU [[^]certification]
- EU Cyber Resilience Act (EU) 2024/2847: designed to meet Annex I, conformity from 11 December 2027
- Australia / New Zealand: RCM - ACMA radiocommunications (AS/NZS 4268), EMC (AS/NZS CISPR 32) and EME [[^]certification]
- Radio module (ESP32-S3): CE, FCC, IC, SRRC, TELEC & BQB certified
- Green certification: REACH/RoHS
- Security support period: 7 years from production date (end date on the label)

Supplied Accessories

- 1x USB Type-C to USB Type-A cable
- 1x RJ12 cable
- 1x Dual RJ12 to pin adapter

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[[^]current]: Current requirements dependant on operating modes and environment.

[[^]usbpower]: USB Type-C port does not supply power.

[[^]rs485power]: RS-485 (parallel RJ12 6P6 ports) supplies power & data to **interfaceNode**.

[[^]jumpers]: Termination and duplex are both set via jumpers internal to the enclosure, not configurable from the web interface - changing either requires opening the unit. For duplex specifically: the jumper selects between the receiver always being enabled (full duplex) or tied to the transmit-enable line (half duplex). The firmware drives a single direction-control signal in either case - the jumper, not the firmware, determines which duplex mode results.

[[^]certification]: Final certification to be completed.

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